Today’s Agenda

- Image Degradation and Restoration

Linear, Position-Invariant Degradations – Noise Free Case

$$g(x,y) = H[f(x,y)]$$

Linearity $H[af_1(x,y)+bf_2(x,y)] = aH[f_1(x,y)]+bH[f_2(x,y)]$

Position/space invariant $H[f(x-\alpha,y-\beta)] = g(x-\alpha,y-\beta)$

$\Downarrow$

H does not depend on the location (x,y), only represented by the input and output

Impulse Response for Linear H

$$f(x,y)=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(\alpha,\beta)\delta(x-\alpha,y-\beta)\,d\alpha d\beta$$

Sifting property

$$g(x,y)=H[f(x,y)]$$

$$g(x,y)=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(\alpha,\beta)H[\delta(x-\alpha,y-\beta)]\,d\alpha d\beta$$

Impulse Response for Linear H

$$g(x,y)=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(\alpha,\beta)H[\delta(x-\alpha,y-\beta)]d\alpha d\beta$$

Impulse response $h(x,\alpha,y,\beta)=H[\delta(x-\alpha,y-\beta)]$

$$g(x,y)=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(\alpha,\beta)h(x,\alpha,y,\beta)d\alpha d\beta$$

If H is position invariant

$$g(x,y)=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(\alpha,\beta)h(x-\alpha,y-\beta)d\alpha d\beta$$

convolution

Image Degradations

$$g(x,y) = H[f(x,y)] + \eta(x,y)$$

$$g(x,y) = \boxed{h(x,y) \otimes f(x,y)} + \eta(x,y)$$

Degradation VS Restoration

$$g(x,y) = H[f(x,y)] + \eta(x,y)$$

Note: a linear, position invariant degradation system with additive noise can be modeled as the convolution of the degradation function with the image plus the additive noise.

$$g(x,y) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(\alpha,\beta)h(x-\alpha, y-\beta)\,d\alpha\, d\beta + \eta(x,y)$$

$$f(x,y) \underset{\text{Restoration (deconvolution)}}{\overset{\text{Degradation (convolution)}}{\rightleftarrows}} g(x,y)$$

Estimate the Degradation Function

- Observation
- Experimentation
- Mathematical modeling

Estimation by Modeling – Motion Blur

Constant velocity along x and y direction:

$$x_0(t) = \frac{at}{T} \qquad y_0(t) = \frac{bt}{T}$$

a b

FIGURE 5.26 (a) Original image. (b) Result of blurring using the function in Eq. (5.6-11) with $a = b = 0.1$ and $T = 1$.

Estimation by Modeling – Cont.

An example of motion blur

$$g(x, y) = \int_0^T f[x - x_0(t), y - y_0(t)]dt$$

Motion in both x and y direction during acquisition

Estimation by Modeling – Cont.

An example of motion blur

$$G(u,v) = F(u,v)\int_0^T e^{-j2\pi[ux_0(t)+vy_0(t)]}dt$$

$$\Rightarrow \quad H(u,v) = \int_0^T e^{-j2\pi[ux_0(t)+vy_0(t)]}dt$$

Estimation by Modeling – Example

Constant velocity along x and y direction:

$$x_0(t) = at/T \qquad y_0(t) = bt/T$$

What is $H(u,v)$?

$$H(u,v) = T\,\frac{\sin[\pi(ua+vb)]}{\pi(ua+vb)}\,e^{-j\pi(ua+vb)}$$

Image Restoration

Given the degradation system **H** and the input image **G**, recover the original image **F**

- Inverse filtering
- Wiener filtering

Inverse Filtering

Ideally:

$$G(u,v) = H(u,v)F(u,v)$$

$$\hat{F}(u,v) = \frac{G(u,v)}{H(u,v)}$$

In practice:

$$G(u,v) = H(u,v)F(u,v) + N(u,v)$$

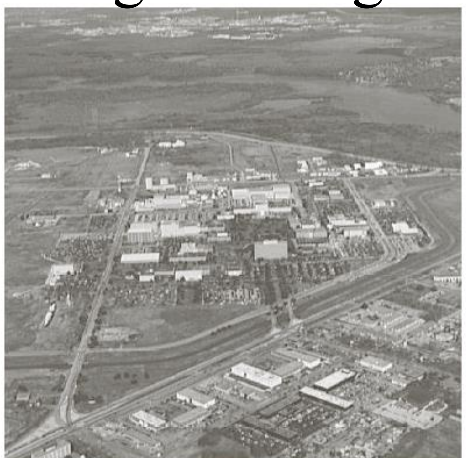
$$\hat{F}(u,v) = F(u,v) + \frac{N(u,v)}{H(u,v)}$$

Limiting the analysis to frequencies near the origin (0,0)

Low pass filtering

An Example of Inverse Filtering

Original image

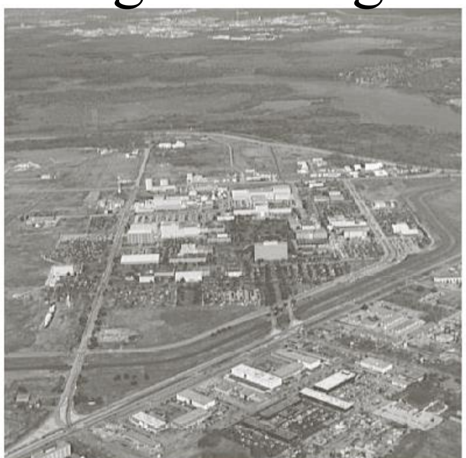


Degraded image

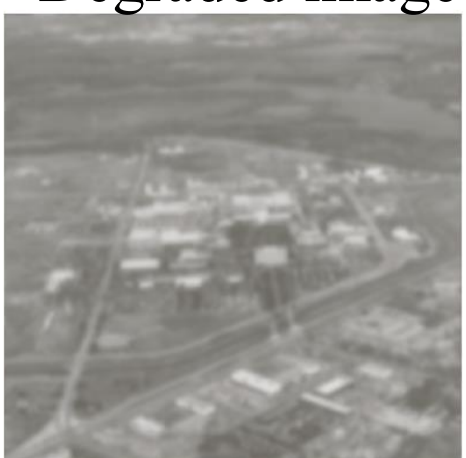
$$\frac{G(u,v)}{H(u,v)}$$

An Example of Inverse Filtering (Cont.)

Original image



Degraded image



$$\frac{G(u,v)}{H(u,v)}$$

Full filter

Cutoff radius = 40

Cutoff radius = 70

Cutoff radius = 85

Minimum Mean Square Error (Wiener) Filtering

Assumptions:

- Noise and image are uncorrelated
- Noise has zero mean

Original

Inverse filtering with cutoff = 70

Wiener filtering

The Formulation

Minimize mean squared error: $e^2 = E\{(f - \hat{f})^2\}$

$$\hat{F}(u,v) = \boxed{\left[\frac{1}{H(u,v)} \frac{|H(u,v)|^2}{|H(u,v)|^2 + |N(u,v)|^2 / |F(u,v)|^2} \right]} G(u,v)$$

Least square error filter

$|N(u,v)|^2$ is the power spectrum of noise

$|F(u,v)|^2$ is the power spectrum of undegraded image

The Formulation (Cont.)

Signal to noise ratio: the metric to evaluate the restoration performance

Frequency domain: $$SNR = \frac{\sum_{u=0}^{M-1}\sum_{v=0}^{N-1}\left|F(u,v)\right|^2}{\sum_{u=0}^{M-1}\sum_{v=0}^{N-1}\left|N(u,v)\right|^2}$$

Spatial domain: $$SNR = \frac{\sum_{u=0}^{M-1}\sum_{v=0}^{N-1}\hat{f}(x,y)^2}{\sum_{u=0}^{M-1}\sum_{v=0}^{N-1}[\hat{f}(x,y)-f(x,y)]^2}$$

The Formulation

$$\hat{F}(u,v) = \left[\frac{1}{H(u,v)} \frac{|H(u,v)|^2}{|H(u,v)|^2 + |N(u,v)|^2 / |F(u,v)|^2} \right] G(u,v)$$

An approximation

$$\hat{F}(u,v) = \left[\frac{1}{H(u,v)} \frac{|H(u,v)|^2}{|H(u,v)|^2 + K} \right] G(u,v)$$

Example 2 – Motion Blur + Additive Noise

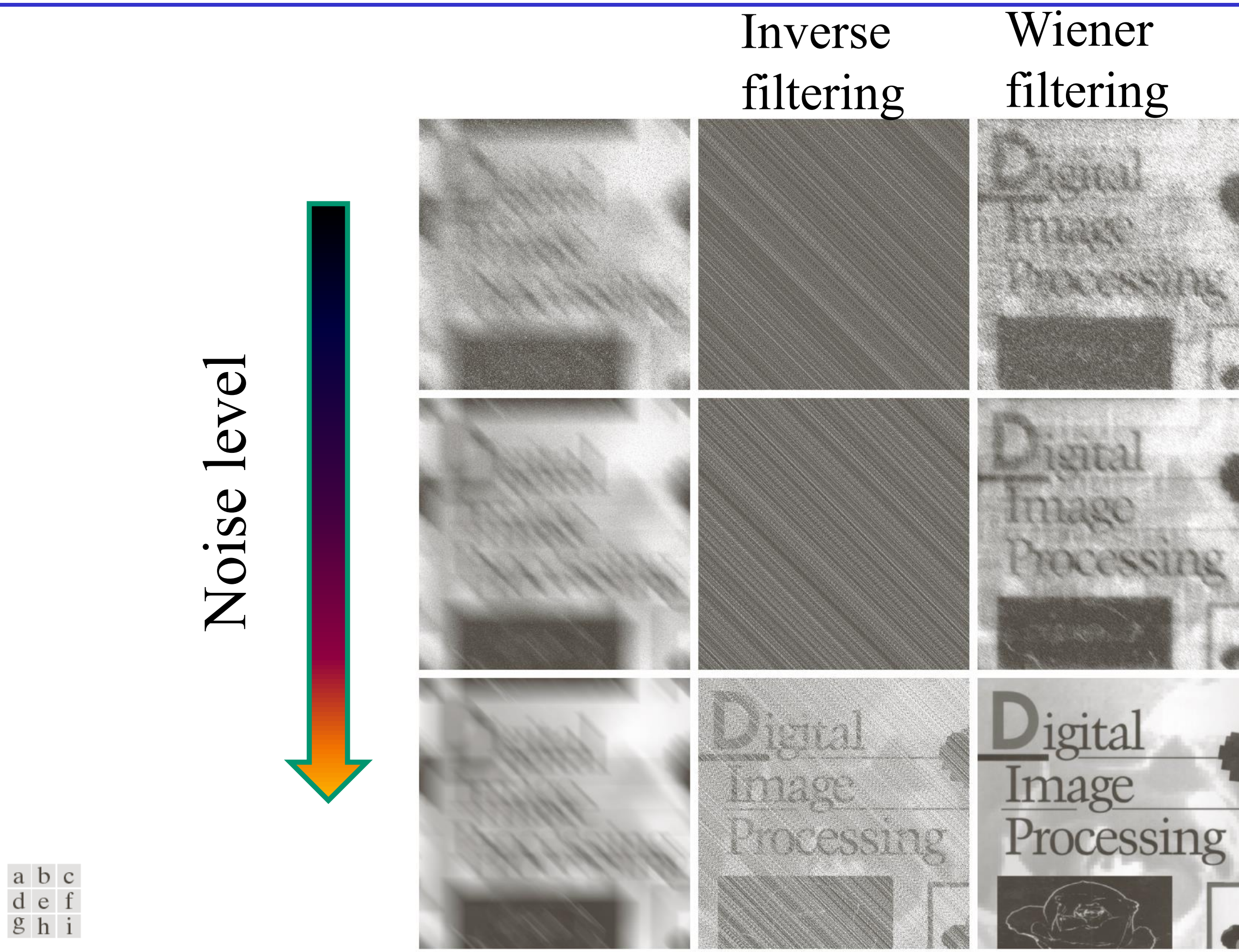


FIGURE 5.29 (a) 8-bit image corrupted by motion blur and additive noise. (b) Result of inverse filtering. (c) Result of Wiener filtering. (d)–(f) Same sequence, but with noise variance one order of magnitude less. (g)–(i) Same sequence, but noise variance reduced by five orders of magnitude from (a). Note in (h) how the deblurred image is quite visible through a "curtain" of noise.

Constrained Least Square Filtering

Assumption: the mean and variance of the noise is known

Matrix-vector representation: $\mathbf{g} = \mathbf{H}\mathbf{f} + \boldsymbol{\eta}$

Alleviate the noise sensitivity by minimizing

$$C = \sum_{u=0}^{M-1}\sum_{v=0}^{N-1}[\nabla^2 f(x,y)]^2$$

The image is smooth

subject to

$$\left\|\mathbf{g} - \mathbf{H}\hat{\mathbf{f}}\right\|^2 = \left\|\boldsymbol{\eta}\right\|^2$$

Constrained Least Square Filtering

Frequency domain solution:

$$\hat{F}(u,v) = \left[\frac{H^*(u,v)}{|H(u,v)|^2 + \gamma |P(u,v)|^2} \right] G(u,v)$$

Fourier transform of Laplacian $p(x,y) = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$

Parameter to satisfy the constraint

Example

CLSF

Wiener

Optimize γ

Residual: $$\mathbf{r} = \mathbf{g} - \mathbf{H}\hat{\mathbf{f}}$$

$$\phi(\gamma) = \|\mathbf{r}\|^2$$ ➡ A function monotonically increasing of γ

Adjust γ to satisfy the constraint:

$$\|\mathbf{r}\|^2 = \|\eta\|^2 \pm a$$ → Accuracy factor

Optimize γ

1. Specify an initial γ
2. Compute $\phi(\gamma) = \|\mathbf{r}\|^2$
3. If $\|\mathbf{r}\|^2 = \|\eta\|^2 \pm a$, stop or adjust γ accordingly

How to Calculate $\|\eta\|^2$

$$\|\eta\|^2 = MN\left({\sigma_\eta}^2 + {m_\eta}^2\right)$$

Example

Iteratively search for optimal parameter γ

a b

FIGURE 5.31 (a) Iteratively determined constrained least squares restoration of Fig. 5.16(b), using correct noise parameters. (b) Result obtained with wrong noise parameters.

Reading Assignments

Chapter 5.10 – 5.11